EDA ON AUTOMOTIVE MARKET TRENDS USING CARS93 DATASET

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DATA NAME:car_price_dataset(from kaggle)

Purpose of the EDA

The primary purpose of this exploratory data analysis is to systematically inspect, clean, and visualize the car_price_dataset to understand underlying data patterns, evaluate data quality, and handle missing values. By utilizing descriptive statistics and visual storytelling (such as distributions and box plots,), this project aims to uncover meaningful insights into how vehicle specifications, brand positioning, and transmission availability impact market pricing and fuel economy

Core Questions Selected for Analysis

Q1: How are car retail prices distributed across the market, and do they cluster heavily toward affordable segments or stretch into luxury tiers?
Q2: What is the direct relationship between vehicle horsepower, engine size, and its final market price?
Q3: Does the availability of a manual transmission option create a noticeable difference in vehicle pricing tiers compared to automatic-only models?

DATA SET OVERVIEW

Total Rows :93
Total Columns :27
Dataset Format : Comma-Separated Values (CSV), loaded dynamically via Python pandas from open-source repositories.

Key Feature Descriptions

The dataset contains a mix of categorical, ordinal, and numerical attributes describing automotive specifications:

Data quality and Cleaning

1. Missing Values Inspection and Treatment

Inspection: Checked all 27 columns for null entries using dataset summary checks (df.isnull().sum()).no missing values were identified exclusively in auxiliary space attributes

2. Duplicate Rows Check

Inspection: Evaluated the entire dataframe for exact duplicate entries using df.duplicated().sum().

3.Outlier Assessment

Inspection: Examined numerical spreads in Price, Horsepower, and EngineSize to spot unusual records.

Treatment Decision: High-end price records (such as luxury models exceeding $50K) and high-performance cars were retained. These represent legitimate market data rather than data-entry errors, adhering to the guideline of treating outliers only when strictly justified.

4. Dataset Size Before and After Cleaning

Before Cleaning: 93 rows and 27 columns.

After Cleaning: 93 rows and 27 columns

Descriptive Analysis

1. Numerical Variables Summary

Price (in $1,000s):

Range: $7.4K to $61.9K

Mean: ~$19.6K | Median: ~$17.7K

Observation: The mean price sits higher than the median, confirming a right-skewed distribution where a tail of high-end luxury models pulls the average upward.

Horsepower (HP):

Range: 55 HP to 300 HP

Mean: ~143.8 HP

Observation: Highlights a broad performance spectrum, spanning from economy commuter cars to high-output sports models.

EngineSize (Liters):

Range: 1.0L to 5.7L

Mean: ~2.66 Liters

Observation: Most vehicles cluster in the 2.0L to 3.0L range, typical for standard passenger cars.

MPG.city & MPG.highway:

City MPG Mean: ~22.3 MPG | Highway MPG Mean: ~26.8 MPG

Observation: Demonstrates expected fuel efficiency differences